

PARTS LIST

Semiconductors:

IC1	- AT89C51 microcontroller
IC2	- HT12D decoder
IC3	- 74LS21 four input AND gate
IC4	- 7805, 5V regulator
IC5	- HT12E encoder
T1	- BC337 npn transistor
T2	- BC547 npn transistor
D1-D5	- 1N4007 rectifier diode
D6-D13	- 1N4148 switching diode
LED1-LED3	- 5mm LED
LCD	- 2-line and 16-character LCD
	- 433MHz ASK transmitter-receiver module

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 10-kilo-ohm
R2, R3, R7	- 1-kilo-ohm
R4	- 3.9-kilo-ohm
R5	- 47-kilo-ohm
R6, R8	- 330-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1, C2	- 33pF ceramic disk
C3	- 1 μ F, 16V electrolytic
C4	- 1000 μ F, 25V electrolytic
C5	- 0.1 μ F ceramic disk

Miscellaneous:

X1	- 230V AC primary to 12V, 500mA secondary transformer
X _{TAL}	- 12MHz crystal
S1, S5, S10-S14	- Push-to-on tactile switch
S6, S7	- SPDT switch
S8	- 4PDT toggle switch
S9	- SPST toggle switch
RL1	- 12V, 1 C/O relay

pull down TE pin low, that address and data are serially transmitted through D_{OUT} pin 17 of HT12E. Here the address is already set to '00h'. Now as any key is pressed, respective data pin goes low and TE pin is grounded. Status of data along with 8-bit address is serially transmitted through ASK transmitter. ASK transmitter modulates these data signal with 433MHz carrier and transmits it through antenna. Table II shows the status of data input pins and the codes transmitted for a particular switch.

Receiver circuit. Receiver circuit is included in Fig. 1. All the address pins of decoder HT12D (IC2) are connected to ground to set same address '00h' as on the transmitter. Data output pins D8 through D11 are connected to microcontroller (IC1) through contacts of 4PDT switch S8. Also, all four outputs are given as input to 4-input

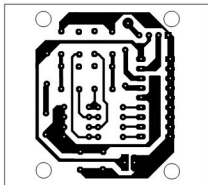


Fig. 6: An actual-size, single-side PCB for RF transmitter

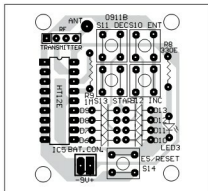


Fig. 7: Component layout for the PCB in Fig. 6

AND gate. The output of AND gate is connected to reset input pin 9 of microcontroller AT89C51 of main circuit. VT pin of HT12D drives LED2 through transistor T1 for indication of data reception. A 51-kilo-ohm resistor is connected between oscillator pins 15 and 16 of decoder HT12D. Output signal of ASK receiver (RX1) is connected with D_{IN} pin 14 of HT12D.

Operation. ASK receiver receives 433MHz carrier signal and demodulates the address and data before giving it to decoder HT12D. HT12D receives these bits serially through D_{IN} pin 14 and compares the address. If received address and set address match, the data will be latched on pins D8 through D11 of decoder HT12D. The VT pin 17 becomes high momentarily as data is latched. This is indicated by LED2. If AD11 pin goes low by pressing start switch (S1) of the transmitter side, only D11 pin 13 at the receiver side becomes low and all other pins become high. Since D11 pin is connected

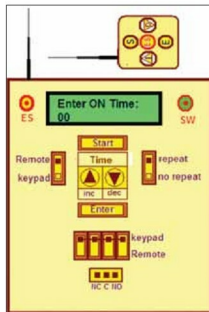


Fig. 8: Proposed arrangement for remote control and front panel for industrial on/off timer

to port pin P1.0 of microcontroller AT89C51, it gives the start signal. Data is received similarly for D8, D9 and D10. When ES key is pressed all data pins D8 through D11 are high. So at the receiver side all the pins become high. That makes the output of AND gate high, which is given to the reset pin 9 of microcontroller.

Because there is a latch action at D8 through D11 outputs, all A through D inputs to AND gate remain high till any other key is pressed at the transmitter. This will continuously apply reset signal to microcontroller AT89C51. To remove reset condition, one SPST (S9) is used in between to make and break the direct connection. Open switch S9 will break the connection and reset pin will not get high logic from the AND gate. Now, after starting the operation, to activate ES from remote, press the switch again to make the connection. Fig. 8 shows the proposed arrangement for front panel and remote control for industrial on/off timer.

Software

The program is written in 'C' language and compiled through Keil μ Vision4 compiler. The program is well-commented and easy to understand. Com-